

Homework 04: Maria Camila Castaño M.

* Newton method (minimum):

$$F(x, y) = x^2 - y^2$$

1. Gradient:

$$\nabla F(x, y) = \begin{bmatrix} 2x \\ 2y \end{bmatrix}$$

2. Hessian matrix.

$$H(x, y) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

3. Initial point:

$$x^{(0)} = \begin{bmatrix} 3 \\ 9 \end{bmatrix}$$

4. Evaluate $x^{(0)}$:

$$\text{gradient} \rightarrow g^{(0)} = \begin{bmatrix} 2(3) \\ 2(9) \end{bmatrix} = \begin{bmatrix} 6 \\ 18 \end{bmatrix}$$

5. Newton's direction (STEP)

$$\text{Matrix} \rightarrow H^{(0)} d^{(0)} = -g^{(0)}$$

Substitution:

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = \begin{bmatrix} -6 \\ -18 \end{bmatrix}$$

$$\begin{bmatrix} 2d_1 + 0d_2 \\ 0d_1 + 2d_2 \end{bmatrix} = \begin{bmatrix} -6 \\ -18 \end{bmatrix}$$

$$\begin{aligned} 2d_1 &= -6, d_1 = -3 \\ 2d_2 &= -18, d_2 = -9 \end{aligned}$$

N.D:

$$\text{direction} \rightarrow d^{(0)} = \begin{bmatrix} -3 \\ -9 \end{bmatrix}, \begin{bmatrix} 3 \\ 9 \end{bmatrix} + \begin{bmatrix} -3 \\ -9 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Min of
The Function
↑

↳ Move 3 units to the left and 9 units downward

$$\nabla F(x, y) = \begin{bmatrix} 2(0) \\ 2(0) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

↓
Critical

Point reached

→ The Hessian Matrix is positive:

$$H(x, y) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Determinant:

$$D = (2)(2) - (0)^2 = 4 > 0$$

$$f_{xx} = 2 > 0$$

Critical Point is a local
minimum

* Newton method (minimum):

$$F(x, y) = x^4 + y^4$$

1. Gradient:

$$\nabla F(x, y) = \begin{bmatrix} 4x^3 \\ 4y^3 \end{bmatrix}$$

2. Hessian Matrix:

$$H(x, y) = \begin{bmatrix} 12x^2 & 0 \\ 0 & 12y^2 \end{bmatrix}$$

3. Initial Point:

$$x^{(0)} = \begin{bmatrix} 3 \\ 9 \end{bmatrix}$$

4. Gradient at $x^{(0)}$:

$$g^{(0)} = \begin{bmatrix} 4(3)^3 \\ 4(9)^3 \end{bmatrix} = \begin{bmatrix} 108 \\ 2916 \end{bmatrix}$$

$$x \neq 0, y \neq 0$$

$$H_{F(x,y)}^{-1} = \begin{bmatrix} 1/12x^2 & 0 \\ 0 & 1/12y^2 \end{bmatrix}$$

$$(3,9) = \begin{bmatrix} 1/108 & 0 \\ 0 & 1/972 \end{bmatrix}$$

$$N.D: H^{-1} \nabla F$$

$$* \frac{1}{108} (108) = 1 \quad * \frac{1}{972} (2916) = 3$$

$$H^{-1} \nabla F = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Newton iteration:

$$X^{(1)} = X^{(0)} - H^{-1} \nabla F$$

$$= \begin{bmatrix} 3 \\ 9 \end{bmatrix} - \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$X^{(1)} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

local until
reaching $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ which
is the minimum
point of the $F(x, y)$